

TAMBACOUNDA, Senegal

WMO: 616870

Lat: 13.737N

Lon: 13.653W

Elev: 49

StdP: 100.74

Time Zone: 0.00 (E00)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	17.6	18.8	-5.8	2.3	28.0	-3.8	2.8	26.8	9.9	28.1	8.9	29.2	2.5	40	0.427

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	12.7	42.2	22.6	41.3	22.4	40.5	22.3	28.5	34.6	27.8	33.3	27.3	32.7	3.3	60

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.1	22.9	31.7	26.4	22.0	30.9	26.0	21.5	30.5	92.3	34.5	89.2	33.3	86.8	32.8	32.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.9	6.8	6.0	14.8	44.1	1.4	1.1	13.7	44.8	12.9	45.5	12.1	46.0	11.1	46.8		
			7.4	29.9	1.1	1.1	6.6	30.7	6.0	31.3	5.3	31.9	4.5	32.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	30.0	26.9	29.2	32.0	33.8	34.0	32.1	29.1	27.9	28.1	29.8	29.4	27.4
	DBStd	2.86	1.98	2.11	1.76	1.35	1.34	2.09	1.68	1.33	1.29	1.37	1.43	1.76
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	7294	523	537	681	715	745	664	593	556	544	614	583	541
	CDD18.3	4253	264	304	422	465	487	414	335	297	294	356	333	282
	CDH23.3	57681	3119	4073	6419	7529	7921	6249	4148	3153	3157	4358	4207	3348
	CDH26.7	34263	1742	2452	4170	5169	5459	3944	1986	1237	1359	2293	2527	1926
Wind	WSAvg	3.1	3.7	3.7	3.3	3.0	3.1	3.4	3.0	2.8	2.3	2.3	2.6	3.4
Precipitation	PrecAvg	773	0	1	0	1	18	94	167	223	189	60	1	0
	PrecMax	1255	1	24	4	42	106	217	336	476	441	207	12	8
	PrecMin	440	0	0	0	0	0	7	6	42	55	0	0	0
	PrecStd	197	0	3	1	6	26	47	64	100	89	48	3	2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	38.2	40.7	42.2	43.3	43.7	41.8	36.9	34.4	35.1	38.8	39.6	38.9
		MCWB	19.8	20.5	21.3	22.2	24.3	25.3	26.9	27.5	27.4	24.6	21.8	19.3
	2%	DB	36.9	39.2	41.2	42.2	42.1	40.1	35.2	33.2	34.1	37.2	38.4	37.8
		MCWB	19.0	20.3	20.9	22.2	23.3	24.9	26.4	26.8	27.3	25.5	21.9	19.5
	5%	DB	35.8	38.0	40.2	41.5	41.1	38.9	34.2	32.4	33.2	36.1	37.7	36.5
		MCWB	18.6	19.8	20.8	22.2	23.3	25.0	26.2	26.4	27.0	26.0	21.7	19.3
	10%	DB	34.4	36.9	39.2	40.5	40.1	37.6	33.2	31.8	32.3	35.1	36.8	35.1
		MCWB	17.9	19.2	20.5	21.9	23.4	24.8	26.1	26.3	26.7	26.2	21.6	18.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	23.1	24.2	24.7	26.6	28.4	29.7	29.2	28.7	28.6	28.9	27.6	23.5
		MCDB	34.5	36.7	38.3	39.5	40.4	38.5	34.4	33.2	32.9	34.6	35.7	34.2
	2%	WB	21.3	22.4	23.6	25.1	26.7	27.7	27.9	27.7	28.0	28.0	26.1	21.7
		MCDB	33.9	35.6	36.7	38.6	37.9	35.7	32.6	31.6	32.4	34.0	33.4	33.9
	5%	WB	19.8	21.2	22.6	24.2	25.7	26.6	27.1	27.0	27.5	27.5	25.2	20.7
		MCDB	33.0	35.1	35.8	37.5	36.6	34.3	32.1	31.0	31.9	33.3	32.0	32.8
	10%	WB	18.6	19.9	21.7	23.3	24.9	26.0	26.5	26.6	27.0	27.0	24.3	19.7
		MCDB	32.5	34.6	35.6	36.4	35.8	33.6	31.5	30.5	31.3	32.7	30.7	32.4
Mean Daily Temperature Range	5% DB	MDBR	15.0	15.0	14.6	14.0	12.7	10.6	8.2	7.6	8.7	11.2	15.4	15.3
		MCDBR	16.9	16.4	15.8	15.0	14.1	12.1	9.2	8.5	9.7	12.8	16.7	17.5
		MCWBR	7.6	7.2	6.4	5.4	4.6	3.9	3.3	3.7	4.4	4.0	5.4	6.8
	5% WB	MCDBR	15.6	15.3	14.5	13.7	12.3	10.8	8.7	8.1	9.3	10.9	13.8	15.8
		MCWBR	7.7	7.6	6.6	5.5	4.8	4.3	3.9	3.9	4.5	4.4	5.1	7.2
Clear-Sky Solar Irradiance	taub	0.517	0.564	0.671	0.670	0.701	0.706	0.622	0.562	0.561	0.597	0.531	0.488	
	taud	1.912	1.800	1.587	1.636	1.596	1.653	1.847	1.999	1.977	1.878	1.957	2.026	
	Ebn at Noon	778	761	693	693	660	648	708	762	765	726	761	793	
	Edh at Noon	189	220	278	264	270	251	208	181	185	200	179	165	
All-Sky Solar Radiation	RadAvg	5.32	5.92	6.61	6.95	6.74	6.20	5.58	5.31	5.45	5.61	5.35	5.11	
	RadStd	0.29	0.27	0.24	0.23	0.27	0.20	0.22	0.21	0.25	0.15	0.22	0.21	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (0 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page